

MECH9325 Acoustics Toolkit

Formula & Use-Case Reference

A calculator-by-calculator summary of every formula in the AcousticsApp toolkit (63 calculators), grouped exactly as they appear in the app's menu. Each entry gives the core equation(s) and, where the calculator answers a distinctive exam-question pattern, a one-line use case. Symbols follow the toolkit's source code:

L/Lp/Lw/LI are levels in dB; p, I, W are the underlying physical quantities; ρc is the characteristic acoustic impedance of the medium.

Reference constant	Value
p_ref (sound pressure)	2e-5 Pa (20 uPa, air)
W_ref (sound power)	1e-12 W
I_ref (sound intensity)	1e-12 W/m ²
ρc (air, default)	415 rayls (1.21 kg/m ³ x 343 m/s)
c (air, default)	343 m/s
Underwater p_ref	1e-6 Pa (1 uPa); typical ρc (water) = 1.5e6 rayls (1000 x 1500)

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Unit Converter

Unit converter (prefixes & acoustic units)

Convert any acoustic quantity between metric-prefixed or named units (Pa, W, Hz, dB refs, etc).

Generic metric prefixes on Pa, W, Hz, s, m, g, V, rayl:

f=1e-15 p=1e-12 n=1e-9 u=1e-6 m=1e-3 c=1e-2 d=1e-1 da=1e1 h=1e2 k=1e3 M=1e6 G=1e9 T=1e12

Also: bar/atm/psi/mmHg (pressure), in/ft/mile (length), min/hr/day (time), rad/s/rpm (frequency), m²/L/m³ (area/volume), km/h/knot (speed), kg/t (mass), C/K/F (temp, affine).

Levels: Basic Conversions & Point-Source Relations

Levels: SPL <-> pressure

Convert between sound pressure level and RMS pressure.

$L_p = 20 \cdot \log_{10}(p / p_{\text{ref}})$ | $p = p_{\text{ref}} \cdot 10^{(L_p/20)}$
 $p_{\text{ref}} = 2 \text{e-}5 \text{ Pa}$

Levels: Sound power level L_w

Convert between sound power (W) and sound power level (dB).

$L_w = 10 \cdot \log_{10}(W / W_{\text{ref}})$ | $W = W_{\text{ref}} \cdot 10^{(L_w/10)}$
 $W_{\text{ref}} = 1 \text{e-}12 \text{ W}$

Levels: Sound intensity level L_I

Sound intensity level, directly from I or from an RMS pressure.

$L_I = 10 \cdot \log_{10}(I / I_{\text{ref}})$ | $I_{\text{ref}} = 1 \text{e-}12 \text{ W/m}^2$
From pressure: | $I = p^2 / (\rho c)$ | [$\rho c = 415 \text{ rayls}$, air, default]

Levels: Peak <-> RMS & combine tones

Peak-to-RMS pressure, and quadrature (energy) combination of several tone components.

$p_{\text{rms}} = P / \sqrt{2}$
Combine components: | $p_{\text{tot}} = \sqrt{\sum p_i^2}$

Levels: PSD -> RMS pressure

RMS pressure and SPL from a straight-line power-spectral-density segment.

$p_{\text{rms}}^2 = 1/2 (S_1 + S_2)(f_2 - f_1)$ | [trapezoidal rule, S in Pa²/Hz]
 $\text{SPL} = 20 \cdot \log_{10}(p_{\text{rms}} / p_{\text{ref}})$

Levels: Radiated power (point source)

Point-source radiation relations at a distance: give ONE of W, I or peak pressure P, get the rest.

$S = 4 \cdot \pi \cdot r^2 / Q$ | (radiating area; Q: 1 free field, 2 hemisphere, 4 edge, 8 corner)
From P: | $p_{\text{rms}} = P / \sqrt{2}$, | $I = p_{\text{rms}}^2 / (\rho c)$ | From W: | $I = W / S$
 $p_{\text{rms}} = \sqrt{I \cdot \rho c}$ | $W = I \cdot S$
 $\text{SPL} = 20 \cdot \log_{10}(p_{\text{rms}} / 2 \text{e-}5)$ | $\text{SIL} = 10 \cdot \log_{10}(I / 1 \text{e-}12)$ | $L_w = 10 \cdot \log_{10}(W / 1 \text{e-}12)$

Use case: 'A spherical source radiates W watts... find p_{rms}, SPL, I, SIL at distance r.'

Levels: Power into a medium (pipe/interface)

Sound power carried into a second medium (e.g. water) through an anechoically-terminated pipe.

```
p_rms = p_ref * 10^(SPL/20) | [continuous across the interface]
I = p_rms^2 / (rho * c)_medium | W = I * A, | A = pi*d^2/4
Lw = 10*log10(W / 1e-12)
```

Use case: pipe immersed in a lake, perforated to act anechoically; SPL measured at the interface.

Combine & Subtract Sources

Combine: add sound levels

Energy (incoherent) sum of any list of decibel levels.

```
L_tot = 10*log10( sum 10^(Li/10) )
```

Combine: N identical sources

Combined level of N identical incoherent sources, from one source's level.

```
L_tot = L1 + 10*log10(N)
```

Combine: increase from more sources

Level rise when 'added' extra identical sources join N1 existing ones.

```
N2 = N1 + added
dL = 10*log10(N2/N1) | L_new = L1 + dL
```

Combine: error using larger signal only

Fractional error made by ignoring the smaller of two signals (ratio = $p_2/p_1 \leq 1$).

```
p_tot = p1 * sqrt(1 + ratio^2)
error = 1/sqrt(1 + ratio^2) - 1
```

Combine: max sources under a limit

Largest number of identical sources whose combined level stays under a permitted limit.

```
one source: L1 = Ltot - 10*log10(N1)
N <= N1 * 10^((Lmax - Ltot)/10) | [round down]
```

Subtract: remove background / source

Energy-subtract one level (e.g. background noise) from a measured total.

```
L_remaining = 10*log10( 10^(Ltot/10) - 10^(Lremove/10) )
```

Subtract: one of N identical sources

Level of a single source, given the combined level of N identical ones.

```
L1 = Ltot - 10*log10(N)
```

Use case: 'two identical fans give 82 dB together; find the level of one fan.'

Waves

Waves: $c = f \times \lambda$

Solve the basic wave relation for whichever quantity is missing; also reports T, omega, k.

```
c = f * lambda | T = 1/f | omega = 2*pi*f | k = omega/c = 2*pi/lambda
```

Waves: speed of sound from temperature

Speed of sound from air temperature, or (inverse) temperature from a measured speed / travel time.

Forward: | $T_0 = T_c + 273.2$ (K), | $c = \sqrt{\gamma R T_0}$ | [$\gamma=1.4$, $R=287$ J/kg/K]

Inverse (from measured c, or distance/time $c=d/t$): | $T_0 = c^2/(\gamma R)$, | $T_c = T_0 - 273.2$

Use case: an impulse crosses d metres between two probe microphones in time t -> estimate air temperature.

Waves: particle velocity & displacement

Particle velocity, displacement, intensity and SPL of a plane wave from its peak pressure.

$u = P / (\rho c)$ | particle velocity amplitude

$\xi = u / \omega$, | $\omega = 2\pi f$ | displacement amplitude

$I = P^2 / (2 \rho c)$ | intensity

$p_{rms} = P / \sqrt{2}$ | $L_p = 20 \log_{10}(p_{rms} / p_{ref})$

Waves: plane wave u -> pressure -> SPL (air/water)

RMS particle velocity of a plane progressive wave -> pressure -> SPL, in air OR water.

$p = \rho c u$ | $L_p = 20 \log_{10}(p / p_{ref})$

Air: | $\rho=1.21$, $c=343$, | $p_{ref}=2e-5$ Pa (20 uPa)

Water: $\rho=1000$, $c=1500$, $p_{ref}=1e-6$ Pa (1 uPa)

Waves: octave band edges & pipe modes

Lower/upper edges and bandwidth of an octave or 1/3-octave band about a centre frequency.

Octave: | $f_{lower} = f_c/\sqrt{2}$, | $f_{upper} = f_c\sqrt{2}$ | -> %BW = 70.7% (constant)

1/3-octave: | ratio $2^{(1/6)}$ | -> %BW = 23.1% (constant)

Waves: pipe natural frequencies (ends)

Natural (resonant) frequencies of an air column in a pipe, by end condition.

closed-open | (quarter-wave): | $f_n = (2n-1) c / (4L)$, | $n = 1, 2, 3, \dots$

open-open | (half-wave): | $f_n = n c / (2L)$

closed-closed (half-wave): | $f_n = n c / (2L)$

Pressure node: centre ($L/2$) for closed-closed; the open end(s) otherwise

Distance Propagation

Distance: attenuation L2 at new distance

Propagate a measured level out to a new distance, for a point or a line source.

Point (spherical, -6 dB/doubling): | $L_2 = L_1 - 20 \log_{10}(r_2/r_1)$

Line | (cylindrical, -3 dB/doubling): $L_2 = L_1 - 10 \log_{10}(r_2/r_1)$

Distance: solve distance from two levels

Back-solve an unknown source distance y from two levels a known increment dr apart.

Point: $y = dr / (10^{(dL/20)} - 1)$ | Line: $y = dr / (10^{(dL/10)} - 1)$

$dL = L_1 - L_2$ | (L_1 nearer, L_2 farther by dr)

Use case: 'level is L_1 at an unknown distance y; moving dr further gives L_2 ; find y.'

Distance: $L_w \leftrightarrow L_p$ (free field / ground)

Convert between sound power level and SPL at a distance, for point or line sources in various fields.

Point: $L_p = L_w - 20 \log_{10}(r) - 10 \log_{10}(4\pi/Q)$ | [Q: 1 free field, 2 ground, 4 edge, 8 corner]

Line: | $L_p = L_w - 10 \log_{10}(r) - k$ | [k = 8 free field, 5 on ground]

Room Acoustics

Room: Sabine reverberation time

Solve the Sabine equation for whichever ONE of V, S, alpha, T60 is unknown.

$$T60 = 0.161 * V / (\alpha * S) \mid A = \alpha * S \mid (\text{total absorption, m}^2)$$

Room: average absorption coefficient

Area-weighted mean absorption coefficient over several surfaces.

$$\alpha_{\text{bar}} = \sum(\alpha_i * S_i) / \sum(S_i)$$

Room: room constant R

Room constant from average absorption and total surface area.

$$R = \alpha * S / (1 - \alpha)$$

Room: room equation Lp from Lw

SPL at a distance from a source, combining the direct and reverberant fields.

$$L_p = L_w + 10 * \log_{10} \left(\frac{Q}{(4 * \pi * r^2)} + \frac{4}{R} \right)$$

Room: reverberant change (add/remove panels)

Change in reverberant SPL when a patch of absorber is added to or removed from a room.

$$\begin{aligned} A_1 &= 0.161 * V / T60 \mid (\text{existing absorption}) \\ A_{\text{abs}} &= S_{\text{abs}} * \alpha \mid (\text{absorber contribution}) \\ A_2 &= A_1 \pm A_{\text{abs}} \mid (\text{new absorption; + add, - remove}) \\ \Delta L_p &= 10 * \log_{10}(A_1 / A_2) \mid (\text{negative when adding absorber}) \end{aligned}$$

Use case: predict the SPL rise/drop when acoustic panels are added or removed from a room.

Room: plant room (surface treatment)

Reverberant-field level in a plant room with several machines, before/after a surface coating.

$$\begin{aligned} \text{Per octave band: } L_w &= 10 * \log_{10} \left(\sum 10^{(L_{w_machine}/10)} \right) \mid (\text{combine machines}) \\ \text{Before: } \alpha_{\text{bar}} &= \alpha_{\text{base}} \\ \text{After: } \alpha_{\text{bar}} &= [S_{\text{coat}} * \alpha_{\text{coat}} + (S - S_{\text{coat}}) * \alpha_{\text{base}}] / S \\ R_{\text{room}} &= S * \alpha_{\text{bar}} / (1 - \alpha_{\text{bar}}) \mid L_p = L_w + 10 * \log_{10}(4/R_{\text{room}}) \\ \text{Overall dB(A)} &= \text{energy-sum of } (L_p + A\text{-weighting}) \text{ across bands; reduction} = \text{before} - \text{after} \end{aligned}$$

Use case: motors in a plant room; combine their Lw per band, get reverberant SPL, then treat a surface.

Room: reverberation test room (mean-square p)

Reverberant mean-square pressure in a test room, empty vs furnished (exact impedance form).

$$\begin{aligned} A &= 0.161 * V / T60 \mid \alpha = A/S \mid R_{\text{room}} = A/(1 - \alpha) \\ W &= W_{\text{ref}} * 10^{(L_w/10)} \\ \langle p^2 \rangle &= 4 * \rho * c * W / R_{\text{room}} \mid L_p = 10 * \log_{10}(\langle p^2 \rangle / p_{\text{ref}}^2) \end{aligned}$$

Sound Power Measurement

Power: background correction K1

Background-noise correction for a sound power measurement (requires dL > 6 dB).

$$\begin{aligned} dL &= L_{\text{source}} - L_{\text{background}} \\ K_1 &= -10 * \log_{10}(1 - 10^{(-dL/10)}) \mid [\text{negligible when } dL \geq 15 \text{ dB}] \end{aligned}$$

Power: environmental correction K2

Room-reflection correction from the measurement surface area and the room's absorption area.

$$K2 = 10 \cdot \log_{10}(1 + 4 \cdot S/A)$$

Power: sound power level (measured)

Sound power level from a mean measurement-surface SPL, with K1/K2 corrections.

$$L_w = (L_p - K1 - K2) + 10 \cdot \log_{10}(S / S_0), \quad | \quad S_0 = 1 \text{ m}^2$$

Power: Lw from enveloping-surface SPLs

Sound power level from SPLs measured on the partial areas of an enveloping surface (e.g. a cuboid).

$$\begin{aligned} L_w &= 10 \cdot \log_{10}(\sum S_i \cdot 10^{(L_i/10)}) \quad | \quad [S_0 = 1 \text{ m}^2] \\ &= \text{mean}L_p + 10 \cdot \log_{10}(S), \quad | \quad \text{mean}L_p = 10 \cdot \log_{10}((1/S) \sum S_i \cdot 10^{(L_i/10)}) \end{aligned}$$

Use case: octave-band SPLs measured on each face of an imaginary box around a machine.

Power: Lw from free-field band SPLs

Free-field sound power level from measured octave-band SPLs over a hemisphere/measurement surface.

$$\begin{aligned} L_{\text{lin},i} &= L_i - W_i \quad | \quad (\text{remove weighting}) \\ L_p &= 10 \cdot \log_{10}(\sum 10^{(L_{\text{lin},i}/10)}) \\ I &= p_{\text{ref}}^2 \cdot 10^{(L_p/10)} / (\rho c) \quad | \quad W = I \cdot S, \quad | \quad L_w = 10 \cdot \log_{10}(W/W_{\text{ref}}) \end{aligned}$$

Use case: octave-band A-weighted SPLs at several points on a hemisphere around a source (e.g. a drill).

Duct / Pipe

Duct: sound power -> mic voltage

Microphone voltage from a duct's sound power (plane-wave), given the mic's V/Pa sensitivity.

$$\begin{aligned} W &= W_{\text{ref}} \cdot 10^{(L_w/10)} \quad | \quad A = \pi \cdot d^2/4 \quad | \quad I = W/A \\ p_{\text{rms}} &= \sqrt{I \cdot \rho \cdot c} \quad | \quad V = p_{\text{rms}} \cdot 10^{(\text{sens}/20)} \\ \text{First higher-order mode cut-on: } &| \quad f_c = 1.8412 \cdot c / (\pi \cdot d) \end{aligned}$$

Duct: octave-band Lp -> I, W, totals (pipe)

Full per-band chain (p_rms, I, W) plus overall Lp/Lw totals for a long reflection-free pipe.

$$\begin{aligned} p_{\text{rms}} &= p_{\text{ref}} \cdot 10^{(L_p/20)} \quad | \quad I = p_{\text{rms}}^2 / (\rho c) \\ W &= I \cdot A, \quad | \quad A = \pi \cdot d^2/4 \quad | \quad (\text{pipe cross-section}) \\ L_{p_{\text{total}}} &= 10 \cdot \log_{10}(\sum 10^{(L_p/10)}) \quad | \quad L_{w_{\text{total}}} = 10 \cdot \log_{10}(\sum W / W_{\text{ref}}) \end{aligned}$$

Use case: speaker in a long pipe, no reflections; octave-band SPLs given -> find p_rms, I, W per band, totals.

Weighting & Band Workbench

A / B / C Weighting & overall level

Apply A/B/C/Z frequency weighting to a band spectrum and energy-sum to an overall level.

$$\begin{aligned} L_{\text{weighted}} &= 10 \cdot \log_{10}(\sum 10^{((L_i + W_i)/10)}) \\ L_{\text{linear}} (\text{unweighted}) &= 10 \cdot \log_{10}(\sum 10^{(L_i/10)}) \end{aligned}$$

Band Workbench (1/3-oct -> octave)

Combine 1/3-octave band SPLs into octave bands, then an overall (and weighted) level.

$$\begin{aligned} \text{Octave SPL} &= 10 \cdot \log_{10}(\sum 10^{(L_{\text{third}}/10)}) \quad | \quad [\text{over the 3 thirds in that octave}] \\ \text{Overall} &= 10 \cdot \log_{10}(\sum 10^{(L_{\text{oct}}/10)}) \quad | \quad \text{Weighted} = 10 \cdot \log_{10}(\sum 10^{((L_{\text{oct}} + W_{\text{oct}})/10)}) \end{aligned}$$

Time-Averaged Levels (Leq / SEL / Dose)

Leq from levels & durations

Equivalent continuous level (and SEL) from a set of level/duration segments.

$$Leq = 10 \cdot \log_{10} \left(\frac{1}{T} \cdot \sum t_i \cdot 10^{(L_i/10)} \right)$$

$$SEL = Leq + 10 \cdot \log_{10}(T / 1 \text{ s})$$

T defaults to sum(durations); pass a fixed T (e.g. 8 h) for LAeq,8h

Use case: a sound level meter's readings over several time periods -> LAeq,T and LAeq,8h.

Leq from discrete events (pass-bys)

Leq over a reference period from repeated discrete events (e.g. vehicle pass-bys).

$$Leq = 10 \cdot \log_{10} \left(\frac{1}{T} \cdot \sum N_i \cdot t_i \cdot 10^{(L_i/10)} \right)$$

Ni = number of events, ti = single-event duration, T = reference period

Leq time-varying level & percentile LN

Leq and a percentile level LN from a segmented time history, including ramp segments.

Ramp segment: | $L(t) = 10 \cdot \log_{10}(a \cdot t + b) + c$

Leq: integrate $10^{(L(t)/10)}$ dt over all segments, divide by total T, $10 \cdot \log_{10}(.)$

LN: the level exceeded for N% of the total time (e.g. N=10 gives L10)

Use case: a noise event whose level follows a formula (e.g. rising then constant) -> LAeq and LA10.

Noise Dose & max time

Occupational noise dose and LAeq against the OH&S exchange-rate criterion.

Allowed time: | $T_i = T_c / 2^{((L_i - L_c)/q)}$ | [Lc=85 dB(A), q=3 dB, Tc=8 h, defaults]

Dose = $\sum (t_i / T_i)$ | [100% = the limit]

$$LAeq,T = 10 \cdot \log_{10} \left(\frac{1}{T} \sum t_i \cdot 10^{(L_i/10)} \right)$$

$$T_{max} = T_c / 2^{((LAeq,T - L_c)/q)}$$

Max permissible time (steady level)

Maximum daily exposure time for a single steady level, using the OH&S exchange rule.

$$T = T_c / 2^{((L - L_c)/q)} \quad | \quad [L_c=85 \text{ dB(A)}, q=3 \text{ dB}, T_c=8 \text{ h, defaults}]$$

Use case: 'a worker is exposed to L dB(A); what is the maximum permissible exposure time?'

Stats: SEL <-> Leq

Convert between the Leq over an event and its Sound Exposure Level (1-second reference).

$$SEL = Leq + 10 \cdot \log_{10}(T / 1 \text{ s})$$

Hearing Protection

Hearing protector (SLC80)

Protected A-weighted level using the single-number SLC80 (AS/NZS 1269) method.

$$LAeq' = LAeq - SLC80$$

(SLC80 subtracts from the C-weighted level, not the A-weighted one)

Hearing protector (octave-band mean/SD)

A-weighted level at the ear from octave-band mean + standard-deviation attenuation data.

Assumed attenuation per band = mean - k*SD | [k=1 'maximum'/worst-case, Bies & Hansen; k=0 mean; k=2 mean-2SD]

$L_{\text{ear}} = L_p - (\text{mean} - k \cdot \text{SD})$

$L_{\text{ear},A} = L_{\text{ear}} + A\text{-weighting}(\text{freq})$ | $LA = 10 \cdot \log_{10}(\sum 10^{(L_{\text{ear},A}/10)})$

Use case: NAL octave-band muff/plug attenuation data + environment octave SPLs -> max LA at the ear.

Loudness

Loudness: phons -> sones

Convert loudness level (phons, from equal-loudness contours) to loudness (sones).

$S = 2^{((LL - 40)/10)}$ | [valid for LL >= 40 phon]

Loudness: sones -> phons

Convert loudness (sones) back to loudness level (phons).

$LL = 40 + 10 \cdot \log_2(S)$

Speech & Community Noise

Speech: PSIL & voice effort

Speech interference level from octave-band noise, and the resulting required voice effort.

$SIL = \text{mean}(\text{octave band levels})$

Classic SIL: 500/1000/2000/4000 Hz. | Preferred PSIL: 500/1000/2000 Hz

Compare SIL (or the resulting voice level) against distance to judge communication effort

Community: day-night level L_{dn}

Community day-night average level, applying the standard 10 dB night-time penalty.

$L_{dn} = 10 \cdot \log_{10}((1/24)[15 \cdot 10^{(L_{\text{day}}/10)} + 9 \cdot 10^{((L_{\text{night}}+10)/10)}])$

[15 h day, 9 h night, +10 dB night penalty]

Statistics

Percentile level LN (part of the time-varying Leq tool)

Statistical level LN exceeded for N% of a sampled time history.

LN = the level exceeded for N% of the time history

e.g. N=10 -> L10, | N=90 -> L90

Stats: sort values into terms

Assign a set of unlabelled dB values to L1 / Leq / L99 / SEL by their expected relative magnitude.

Sort the given values in DESCENDING order, then assign by physical meaning:

SEL | (largest - normalised to 1 s, energy-concentrated)

L1 | (2nd largest - level exceeded only 1% of the time, near the peak)

Leq | (3rd - the time/energy average, between the extremes)

L99 | (smallest - level exceeded 99% of the time, i.e. the background)

Use case: 'the values X, Y, Z, W were measured for L1, L99, Leq and SEL but not which is which.'

Sound Insulation (Transmission Loss)

Insulation: mass law TL

Normal-incidence transmission loss of a limp panel from the mass law; layers in contact ADD mass.

$TL = 20 \cdot \log_{10}(M \cdot f) - 42.4$ | [M = surface mass, kg/m²; layers/glued panels: M = sum of layer masses]

Use case: 'find the mass-law TL of a Xkg/m² panel at several frequencies.'

Insulation: composite / double-panel TL (TL1+TL2)

TL of panels in series (a composite barrier, e.g. two leaves with an air gap/blanket between).

$\alpha_t = \alpha_1 \cdot \alpha_2 \cdot \dots$ | (transmission coefficients multiply)
 $TL = TL_1 + TL_2 + \dots$ | (so individual TLs simply ADD)

Use case: two panels mounted apart with a blanket between them -> composite TL.

Insulation: interface impedance & coeffs

Reflection/transmission at a normal-incidence interface between two media (e.g. air-water).

$r = z_2/z_1$ | $\alpha_t = 4r/(r+1)^2$ | (power transmitted) | $\alpha_r = ((r-1)/(r+1))^2$ (reflected)
 $T_p = 2z_2/(z_1+z_2)$ | (pressure transmission coeff) | $R_p = (z_2-z_1)/(z_1+z_2)$ | (pressure reflection coeff)
 $TL = -10 \cdot \log_{10}(\alpha_t)$
 $p_{Trans} = T_p \cdot p_0$ | $p_{Interface} = p_{Trans}$ | $u_{Trans} = p_{Trans}/z_2$ | $p_{Ref} = R_p \cdot p_0$

Use case: plane wave normally incident on an interface (e.g. air-water); find transmitted p, u, and tau.

Insulation: TL from coefficient

Transmission loss directly from a known transmission coefficient.

$TL = -10 \cdot \log_{10}(\alpha_t)$

Insulation: panel resonance frequency

Fundamental resonance frequency of a mass-spring (limp) panel.

$f_n = (1/2\pi) \cdot \sqrt{K / M}$ | [K = stiffness/area N/m², M = surface mass kg/m²]

Insulation: panel TL (resonant, mass+stiffness)

TL of a resonant panel (mass-spring model) across stiffness- and mass-controlled regions.

$k = F/\text{deflection}$ (from a static test) | $s' = k/A$ | $m' = \text{density} \cdot \text{thickness}$
 $f_n = (1/2\pi) \cdot \sqrt{s'/m'}$
 $f < f_n$ (stiffness-controlled): | $TL = 20 \cdot \log_{10}(s' / (2 \cdot \rho \cdot c \cdot w))$ |
 $f > f_n$ (mass-controlled): | $TL = 20 \cdot \log_{10}(w \cdot m' / (2 \cdot \rho \cdot c))$ |, | $w = 2\pi \cdot f$

Use case: a panel deflects a known amount under a static force -> find f_n and TL above/below resonance.

Insulation: barrier added to a spectrum (dB(A) before/after)

Effect of inserting a mass-law barrier into an octave-band spectrum: overall dB(A) before/after.

Per band: $TL = 20 \cdot \log_{10}(m' \cdot f) - 42.4$ | $L_{p_after} = L_p - TL$
Overall (weighted) = $10 \cdot \log_{10}(\sum 10^{((L_p + W_i)/10)})$ | [before and after]

Use case: 'a barrier/disc is fitted in a duct/tunnel; find the overall dB(A) before and after.'

Mufflers / Silencers

Muffler: sudden area change

Transmission loss of a sudden pipe cross-section change (reactive muffler element).

$$T_t = 4 S_1 S_2 / (S_1 + S_2)^2 \quad | \quad TL = -10 \log_{10}(T_t)$$

Muffler: simple expansion chamber

TL of a simple (reactive) expansion-chamber muffler at a given frequency.

$$m = S_2 / S_1 \quad | \quad kL = (2\pi f / c) * L$$
$$TL = 10 \log_{10} [\cos^2(kL) + 1/4 (m + 1/m)^2 \sin^2(kL)]$$

Muffler: TL / IL / NR (level difference)

The common level-difference definition underlying TL, insertion loss and noise reduction.

$$TL = L_{w1} - L_{w2} \quad | \quad (\text{transmission loss}) \quad | \quad IL = L_{\text{before}} - L_{\text{after}} \quad | \quad (\text{insertion loss})$$
$$NR = L_{\text{in}} - L_{\text{out}} \quad | \quad (\text{noise reduction})$$